LAKSH BALIRAM TALREJA

Cybersecurity Engineer | Vulnerability Assessment · Detection · MITRE ATT&CK · Python · AWS
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Stamp 1G - Currently authorised to work in Ireland

Professional Summary

Cybersecurity engineer with an MSc in Cybersecurity (Dublin Business School, 2026) and hands-on experience in vulnerability assessment, penetration testing, and detection engineering. Built a Python platform that turns telemetry into CVE/CVSS and MITRE ATT&CK-mapped findings with auditable explanations and reconstructed exploit sequences, plus a secure automated deployment pipeline on AWS.

Technical Skills

Security: Vulnerability assessment, CVE/CVSS triage, penetration testing, MITRE ATT&CK, coordinated vulnerability disclosure, exploitability assessment

Detection & Analysis: Telemetry analysis (Sysmon), SHAP-explained classification, exploit-chain reconstruction, KPI/KRI reporting

Frameworks: ISO/IEC 27001, NIST CSF, NIS/NIS2, ETSI EN 303 645 and IEC 62443 (awareness)

Engineering & Cloud: Python, Flask, AWS, Docker, Terraform, Git, GitHub Actions

Tools: Kali Linux, NVD, OSV, CISA KEV

Professional Experience

Networking & Linux Lab Instructor - D Y Patil International University *Sep 2023 - Dec 2024*

Pune, India | Linux and networking labs, confidential records

- Handled sensitive records confidentially while balancing competing demands from several stakeholders.
- Set up and maintained lab environments for Linux and networking practical sessions across multiple modules, supporting 60+ students.
- Demonstrated hands-on configuration practicals covering IP addressing and core networking fundamentals, guiding students through setup and troubleshooting.
- Troubleshot lab and student technical issues in real time during live practical sessions.
- Wrote lab guides and technical documentation reviewed by academic leadership.
- Coordinated sessions and resources with students, faculty, and administration, escalating conflicts to the appropriate lead.

Cybersecurity Researcher - Cyber3ra *Jun 2022 - Aug 2022 | Remote*

Vulnerability assessment and penetration testing

- Triaged findings by actual exploitability in the target environment instead of relying on CVSS base scores alone.
- Researched and documented component-level attack surfaces across software and OS environments, giving findings both technical and non-technical reviewers could use.
- Carried out structured vulnerability assessments and penetration tests on Linux and Windows environments.
- Wrote vulnerability reports with CVE-referenced risk ratings, exploitability justifications, and prioritised remediation recommendations.
- Worked independently on a remote basis and delivered reports within the internship period.

IoT Developer Intern - D Y Patil University *Jun 2021 - Aug 2021 | Pune, India*

IoT security system

- Gained direct exposure to the point where hardware, firmware, and software meet, which is where product security work happens.
- Designed and built an IoT security system combining hardware sensors, microcontrollers, firmware logic, and a software data pipeline.
- Wrote system architecture documentation and operating procedures for the finished system.

Junior Software Developer - Fortune Cloud *Jan 2020 - Apr 2020 | Pune, India*

Production software development

- Contributed to live production software as part of an established engineering team.
- Worked with professional version control (Git) and followed the team's software release process.

Project Experience

Vulnerability Detection & Risk-Scoring Platform | *Python · Flask · MITRE ATT&CK · CVE/CVSS · SHAP · D3.js · AWS · Sysmon*

- Built a Python/Flask platform that ingests multi-source system telemetry and maps component-level behaviours to CVE/CVSS-referenced risk classifications and MITRE ATT&CK techniques.
- Determined actual exploitability in the specific system context rather than relying on base scores.
- Tracked which processes, binaries, registry dependencies, and software versions were involved in each detection, similar to maintaining a software bill of materials.
- Applied SHAP explainability so each classification carries a transparent, auditable justification.
- Built a forensic module that reconstructs exploit sequences from structured logs and packages the evidence for disclosure or investigation.
- Deployed the platform on AWS.
- Built a real-time KPI/KRI dashboard in D3.js showing vulnerability trends, risk scores, and remediation status.

Secure Cloud Deployment Pipeline | *Terraform · AWS · Ansible · Docker · GitHub Actions CI/CD*

- Built a multi-stage automated deployment pipeline using Terraform, Docker, Ansible, and GitHub Actions on AWS.
- Pinned dependency versions to keep builds reproducible and reduce supply chain risk.
- Applied IAM least-privilege access control across all pipeline stages.
- Wrote architecture documentation, configuration standards, and operational runbooks.

Education

Dublin Business School *Jan 2025 - Jan 2026*

MSc Cybersecurity | Second Class Honours | Dublin, Ireland

Relevant modules: Vulnerability Management & CVE/CVSS Frameworks · Penetration Testing · Coordinated Vulnerability Disclosure · Secure SDLC · Supply Chain Security · ISO/IEC 27001 & ISMS · NIST CSF · NIS/NIS2

D Y Patil International University *Aug 2019 - Jun 2023*

B.Tech. Computer Science Engineering | CGPA: 7.22 / 10 | Pune, India